

NAME

Fingerprints - Fingerprints class

SYNOPSIS

```
use Fingerprints::Fingerprints;

use Fingerprints::Fingerprints qw(:all);
```

DESCRIPTION

Fingerprints class provides the following methods:

new, FoldFingerprintsByBitDensity, FoldFingerprintsBySize, GetFingerprintBitsAsBinaryString, GetFingerprintBitsAsHexadecimalString, GetFingerprintBitsAsRawBinaryString, GetFingerprintsVectorValueIDs, GetFingerprintsVectorValues, IsFingerprintsGenerationSuccessful, SetFingerprintsBitVector, SetFingerprintsVector, SetFingerprintsVectorType, SetMolecule, SetSize, SetType, SetVectorType

Fingerprints class is used as a base class for various specific fingerprint classes such as AtomNeighborhoodsFingerprints, AtomTypesFingerprints, EStateIndicesFingerprints, PathLengthFingerprints, ExtendedConnectivityFingerprints, MACCSKeys and so on. It implements functionality common to fingerprint classes.

Fingerprints class is derived from ObjectProperty base class which provides methods not explicitly defined in Fingerprints or ObjectProperty classes using Perl's AUTOLOAD functionality. These methods are generated on-the-fly for a specified object property:

```
Set<PropertyName>(<PropertyValue>);
$PropertyValue = Get<PropertyName>();
Delete<PropertyName>();
```

Fingerprints class uses FingerprintsBitVector class to provide bits manipulation functionality.

METHODS

new

```
$NewFingerprints = new Fingerprints(%NamesAndValues);
```

Using specified *Fingerprints* property names and values hash, new method creates a new object and returns a reference to newly created Fingerprints object. By default, following properties are initialized:

```
Molecule = '';
Type = '';
VectorType = '';
Size = '';
MinSize = '';
MaxSize = '';
FingerprintsBitVector = '';
FingerprintsVectorType = '';
FingerprintsVector = '';
```

FoldFingerprintsByBitDensity

```
$Fingerprints->FoldFingerprintsByBitDensity($BitDensity);
```

Folds fingerprints by recursively reducing its size by half until bit density is greater than or equal to specified *BitDensity* and returns *Fingerprints*.

FoldFingerprintsBySize

```
$Fingerprints->FoldFingerprintsBySize($Size, [$CheckSizeValue]);
```

Fold fingerprints by recursively reducing its size by half until size is less than or equal to specified *Size* and returns *Fingerprints*. By default, value *Size* is checked to make sure it's:

```
>= MinSize and < Size and IsPowerOfTwo
```

GetFingerprintBitsAsBinaryString

```
$BinaryASCIIString =
  $Fingerprints->GetFingerprintBitsAsBinaryString();
```

Returns fingerprints as a binary ASCII string containing 0s and 1s.

GetFingerprintBitsAsHexadecimalString

```
$HexadecimalString =  
  $Fingerprints->GetFingerprintBitsAsHexadecimalString();
```

Returns fingerprints as a hexadecimal string

GetFingerprintBitsAsRawBinaryString

```
$RawBinaryString =  
  $Fingerprints->GetFingerprintBitsAsRawBinaryString();
```

Returns fingerprints as a raw binary string containing packed bit values for each byte.

GetFingerprintsVectorValueIDs

```
$ValueIDsRef = $Fingerprints->GetFingerprintsVectorValueIDs();  
@ValueIDs = $Fingerprints->GetFingerprintsVectorValueIDs();
```

Returns fingerprints vector value IDs as an array or reference to an array.

GetFingerprintsVectorValues

```
$ValuesRef = $Fingerprints->GetFingerprintsVectorValues();  
@Values = $Fingerprints->GetFingerprintsVectorValues();
```

Returns fingerprints vector values as an array or reference to an array.

IsFingerprintsGenerationSuccessful

```
$Return = $Fingerprints->IsFingerprintsGenerationSuccessful();
```

Returns 1 or 0 based on whether fingerprints were successfully generated.

SetFingerprintsBitVector

```
$Fingerprints->SetFingerprintsBitVector($FingerprintsBitVector);
```

Sets *FingerprintsBitVector* object for *Fingerprints* and returns *Fingerprints*.

SetFingerprintsVector

```
$Fingerprints->SetFingerprintsVector();
```

Sets *FingerprintsVector* object for *Fingerprints* and returns *Fingerprints*.

SetFingerprintsVectorType

```
$Fingerprints->SetFingerprintsVectorType($VectorType);
```

Sets *FingerprintsVector* type for *Fingerprints* and returns *Fingerprints*. Possible *VectorType* values: *OrderedNumericalValues*, *NumericalValues* or *AlphaNumericalValues*.

SetMolecule

```
$Fingerprints->SetMolecule($Molecule);
```

Sets *Molecule* object for *Fingerprints* and returns *Fingerprints*.

SetSize

```
$Fingerprints->SetSize($Size);
```

Sets *Size* of fingerprints and returns *Fingerprints*.

SetType

```
$Fingerprints->SetType($Type);
```

Sets *Type* of fingerprints and returns *Fingerprints*.

SetVectorType

```
$Fingerprints->SetVectorType($Type);
```

Sets *Type* of fingerprints vector and returns *Fingerprints*. Possible *Type* values: *FingerprintsBitVector* or

FingerprintsVector.

AUTHOR

Manish Sud <msud@san.rr.com>

SEE ALSO

FingerprintsStringUtil.pm, AtomNeighborhoodsFingerprints.pm, AtomTypesFingerprints.pm, EStateIndiciesFingerprints.pm, ExtendedConnectivityFingerprints.pm, MACCSKeys.pm, PathLengthFingerprints.pm, TopologicalAtomPairsFingerprints.pm, TopologicalAtomTripletsFingerprints.pm, TopologicalAtomTorsionsFingerprints.pm, TopologicalPharmacophoreAtomPairsFingerprints.pm, TopologicalPharmacophoreAtomTripletsFingerprints.pm

COPYRIGHT

Copyright (C) 2026 Manish Sud. All rights reserved.

This file is part of MayaChemTools.

MayaChemTools is free software; you can redistribute it and/or modify it under the terms of the GNU Lesser General Public License as published by the Free Software Foundation; either version 3 of the License, or (at your option) any later version.